

JSPL EAF Industrial Process Map vs Data Coverage

Phase 27 data-gap analysis. Based on Phase 23 literature (Knutsen, Sjunnesson, Kirschen, Memoli, Duan) and Phase 24–26 empirical findings. Production models frozen.

Process stage	Primary TTT drivers	Currently recorded?	Data gap impact
Scrap charging	Bucket logistics, crane wait, sequence	Scrap tonnes only	HIGH — delays invisible
HM charging	HM mass, temperature, arrival wait	HM tonnes; no temperature/time	MEDIUM–HIGH
DRI feeding	Rate, metallization, temperature	DRI tonnes only	HIGH — quality missing
Melting	Power-on, arc, foam, solid burden	No power-on; no arc V/I	VERY HIGH
Foamy slag	C/O2 timing, FeO, foam index	Proxy only (totals uncertain)	HIGH
Refining	O2 profile, temp endpoint, chemistry	OXY total; no profile/slag lab	HIGH
Tapping / turnaround	EBT, ladle wait, maintenance	TTT end only	HIGH for delays
Abnormal events	Delay codes, trips, electrode break	Inferred from TTT only	CRITICAL

Key conclusion

JSPL currently records **recipe inputs** and a **single outcome (TTT)**, plus an end-of-heat energy total (POWER = EE_KWH) that must not be used as a planning feature. Almost all **process-state and delay-event** measurements are absent. This explains the Phase 26 information ceiling (~3.24 min MAE) and Phase 25 delay-warning recall of only ~22%.

See also presentation_figures/process_data_coverage.png for the stage colour map (green = data, orange = partial, red = gap).